

Multiobject GSE $\pi 0.5$ — Real-Episode Action-Prediction Eval

open-loop / teacher-forced: real observation in $\rightarrow$ predicted joint chunk vs the real demonstrated future

metric	in-distribution	out-of-distribution
frames evaluated	768	768
first-step MAE (33 ms)	1.26°	6.40°
horizon-end MAE (333 ms)	1.31°	8.82°
frames within 2°	90.8%	2.2%
frames within 5°	99.7%	28.1%
displacement r (Pearson)	0.940	0.079
direction agreement (cosine)	0.839	-0.013
velocity match r	0.842	0.162
gripper within 0.1	99.0%	84.0%

Model: pi05_farm_multiobject_gse (step-5999, VLA-GSE). In-dist data = its own training set NoahWeiss/farm_uf850_multiobject (424 eps, 4 tasks). OOD data = held-out NoahWeiss/farm_uf850_bottle (200 eps) — episodes never seen in multiobject training; its 'bottle off box $\rightarrow$ desk' task is an instruction the model never trained on. Metric is single-shot prediction fidelity, not closed-loop success. All figures generated from real frames + real inferences (model/cluster/eval_train_endhorizon.py).

Findings

Does the multiobject GSE $\pi 0.5$ policy predict real demonstrated motion? In-distribution vs. out-of-distribution

****Model under test:**** `pi05\_farm\_multiobject\_gse`, checkpoint `step-5999` (VLA-GSE full fine-tune of $\pi 0.5$ on `NoahWeiss/farm\_uf850\_multiobject`, 424 eps, 4 tasks: bottle/bear/duck/hat $\rightarrow$ box).

TL;DR

We measured open-loop, teacher-forced single-shot prediction fidelity: the model always sees the ****real**** recorded observation (7-DoF joint state + base/wrist cameras + task prompt) and emits a 10-step absolute-joint chunk; we score `pred[k]` against the real recorded `state[t+1+k]` (33 ms/step, horizon end = 333 ms ahead). On its ****own training set**** (Condition A, in-distribution) the policy nearly retraces the demonstrations: 1.31° horizon-end MAE, a flat error-vs-horizon curve, displacement Pearson ****r = 0.940****, and direction cosine ****0.84****. On ****held-out**** episodes (Condition B, OOD: `NoahWeiss/farm\_uf850\_bottle`, 200 eps never in training) the absolute MAE rises "only" 6.7× to 8.82° and 71.7% of frames stay within 10°, which looks like partial success – but every metric that measures whether the prediction tracks the *demonstrated motion* collapses: displacement $r \rightarrow$ ****0.079**** and direction cosine $\rightarrow$ ****−0.013**** (chance), with all per-joint R^2 turning negative. The reconciliation is the finding: within a bounded joint workspace, emitting any plausible in-distribution pose keeps absolute error small, while the motion-structure metrics (which subtract out that static prior) expose that no scene-conditioned visuomotor signal transfers. This is a generalization ****failure**** wearing the costume of partial success.

1. What was measured

****Metric (open-loop / teacher-forced).**** For every sampled frame, the model receives the ****real recorded observation**** at time `t` (7-dim joint state, base + wrist camera frames, task prompt) and outputs a 10-step chunk of absolute joint targets. We compare each predicted step `pred[k]` to the ****real recorded**** state `state[t+1+k]` – this is exactly openpi's training target. The data is 30 fps, so step `k` is `33·(k+1)` ms ahead; the horizon end `pred[9] = state[t+10]` is 333 ms ahead.

This measures ****prediction fidelity****, *not* closed-loop task success. The model never sees its own previous outputs; every prediction is re-grounded on a real observation, so errors do not compound across the metric itself (only within a single 10-step chunk). Sampling: seed 0, 768 random mid-episode frames per condition over 64 episodes, window `[0.25, 0.75]` of each episode (the high-motion middle, where a static pose prior helps *least*), plus dense whole-episode rollouts for the trajectory figures.

****Step clock.**** `pred[k]  $\leftrightarrow$  state[t+1+k]`, `k = 0...9`. First-step = 33 ms, horizon-end = 333 ms.

****The two datasets.****

- ****Condition A (in-distribution, FIT):**** `NoahWeiss/farm\_uf850\_multiobject` – the model's ****own**** training set (424 eps; 4 tasks bottle/bear/duck/hat $\rightarrow$ box).

Findings (cont.)

- **Condition B (out-of-distribution, GENERALIZATION):** `NoahWeiss/farm\_uf850\_bottle` – 200 held-out episodes from a *separate* collection session, never in multiobject training. Two task strings:

1. **"Picking up the bottle and placing it on the box"** – the **same instruction string** as a multiobject training task, but different/held-out episodes (a same-task, cross-collection shift).
2. **"Picking up the bottle off of the box and putting it on the desk"** – a **reverse task** whose instruction the multiobject model never trained on (unambiguous novel-instruction OOD).

Why this is a valid OOD probe. Same UF850 arm, same 7-DoF absolute joint space, same 30 fps, same base+wrists cameras as the training data, so the observation/action interface matches deployment. The never-trained reverse instruction guarantees at least one split is unambiguously novel; the held-out episodes guarantee that even the seen-instruction split tests cross-collection novelty rather than memorized frames. Normalization is handled correctly (see §7).

2. In-distribution results (fit)

On its own training set the policy reproduces the demonstrated motion almost exactly. This is **fit / memorization**, not generalization – and it is the floor against which Condition B is read.

Metric (Condition A)	Value
First-step MAE (33 ms)	1.26°
Horizon-end MAE (333 ms)	1.31°
Chunk-mean MAE	1.25°
Within 2° / 5° / 10°	90.8% / 99.7% / 100%
Displacement Pearson r (333 ms)	0.940
Per-joint displacement R ²	[0.83, 0.93, 0.76, 0.89, 0.87, 0.85] (all +)
Motion-direction sign agreement (per joint)	[0.83, 0.97, 0.88, 0.86, 0.86, 0.88]
Mean 6-vector displacement cosine	0.84 (median 0.93)
Velocity-profile Pearson r	0.842
Gripper within 0.1	99.0%

Error does not compound across the chunk. The error-vs-horizon curve is flat: `[1.26, 1.24, 1.22, 1.21, 1.22, 1.22, 1.24, 1.26, 1.27, 1.31]°` – the 333 ms-ahead step is only 0.05° worse than the first step. This means the 10-step chunk has **uniform per-step prediction accuracy** against the teacher-forced targets (no compounding *in prediction*). It does **not** imply anything about closed-loop execution reliability or task success, which this metric cannot measure. See `figs/horizon_indist.png`.`

The model reproduces motion, not a static held pose. Displacement $r = 0.940$ with all-positive per-joint R², per-joint direction agreement well above the 50% chance baseline, mean displacement cosine 0.84, and velocity $r = 0.842$ together show the policy correctly commands where each joint is heading and roughly how fast. The displacement scatter (`figs/disp_indist.png`)` is a tight diagonal cloud.

Findings (cont.)

****Per-task structure**** (`figs/pertask\_indist.png`). Bottle dominates the sample and fits best; rarer tasks fit slightly worse, roughly in inverse-frequency order, but all stay under 1.6° with 99–100% within 5%:

Task	Horizon-end MAE	Within 5%	n
bottle → box	1.24°	99.8%	528
bear → box	1.42°	99.2%	132
duck → box	1.45°	100%	24
hat → box	1.52°	100%	84

Gripper open/close is reproduced almost perfectly (99.0% within 0.1), consistent with the clean square-wave tracking in the trajectory figures (`figs/traj\_indist\_\*.png`), which show the 10-step-lookahead and next-angle predictions overlaying the real demo across all 6 joints + gripper with only hairline deviation.

3. Out-of-distribution results (generalization)

On held-out episodes the same metric collapses on every motion-structure axis, even as absolute MAE stays deceptively bounded.

Metric	In-distribution (A)	Out-of-distribution (B)
First-step MAE	1.26°	**6.40°**
Horizon-end MAE (333 ms)	1.31°	**8.82°** (6.7×)
Chunk-mean MAE	1.25°	7.59°
Within 2° / 5° / 10°	90.8% / 99.7% / 100%	**2.2% / 28.1% / 71.7%**
Error-vs-horizon shape	FLAT (1.26→1.31)	**GROWS** (6.40→8.82)
Displacement Pearson r	**0.940**	**0.079**
Per-joint displacement R ²	all positive	**[-8.6, -2.7, -9.6, -20.8, -4.2, -11.9] (all negative)
Mean displacement cosine	**0.84** (med 0.93)	**−0.013** (med −0.037)
Direction sign agreement (per joint)	[0.83...0.88]	**[0.47, 0.49, 0.45, 0.52, 0.52, 0.54] (~chance)
Velocity-profile r	0.842	0.162
Gripper within 0.1	99.0%	84.0%

OOD error-vs-horizon: `[6.40, 6.64, 6.91, 7.17, 7.43, 7.71, 7.99, 8.28, 8.56, 8.82]°`
(`figs/horizon\_ood.png`).

The central insight: absolute MAE understates the failure

The "only 6.7× / 71.7% within 10°" headline is misleading. The reconciliation:

Findings (cont.)

- **Absolute MAE is flattered by a bounded workspace prior.** Because the actions are *absolute joint targets*, a model that merely emits a plausible in-workspace pose lands within $\sim 10^\circ$ of any realistic UF850 reach pose by geometry alone. So bounded absolute error reflects the workspace, not tracking skill. *(This workspace-prior mechanism is the best-supported interpretation of the data – it is consistent with the displacement/direction collapse and the disp_ood scatter clustering near the origin; we did not measure an explicit static-pose baseline MAE, so it is an interpretive hypothesis, not a directly measured artifact.)*

- **The prior-free metrics expose the collapse.** Displacement Pearson r falls $0.940 \rightarrow 0.079$ and the 6-vector direction cosine falls $0.84 \rightarrow -0.013$ – exactly chance. The model does not even reliably know which *way* each joint should move (per-joint sign agreement ~ 0.50).

- **The decisive evidence is the all-negative per-joint R^2** $[-8.6, -2.7, -9.6, -20.8, -4.2, -11.9]$. A negative R^2 means the model's commanded displacement is a *worse* predictor of the real demonstrated displacement than a constant "no-motion" baseline. The predictions are *uncorrelated* with the true motion ($r = 0.079$, chance) and their magnitude/offset is wrong enough to score below the mean baseline – this is the signature of emitting a near-static prior trajectory against varying targets, *not* anti-correlation. (The negative magnitudes themselves, e.g. -20.8 on j_4 , are variance-sensitive and should be read as "far below the mean baseline," not as calibrated effect sizes.)

- **The horizon curve flips meaning.** In-dist flat = the motion was actually captured; OOD growing = the model has no correct velocity/direction to extend, so it drifts further from truth the longer it extrapolates. Velocity match drops $0.842 \rightarrow 0.162$.

Verdict: low MAE + chance direction = the model outputs a plausible-looking but scene-blind pose. The 71.7%-within- 10° number must always be read alongside the motion-structure collapse; it is not 71.7% useful predictions.

4. Instruction-novelty split

Splitting Condition B by task separates cross-collection episode/visual novelty from instruction novelty (`figs/pertask_ood.png`):

OOD task	Horizon-end MAE	Within 5%	Direction cosine	n	
--- --- --- --- ---					
"bottle → box" – <i>seen instruction</i>	held-out episodes	9.17°	35.2%	$+0.13$	420
"bottle off box → desk" – <i>never-trained instruction</i>		8.40°	19.5%	-0.19	348

Two observations:

- The seen instruction buys only a faint directional prior.** The familiar prompt string keeps direction cosine weakly positive ($+0.13$) – a residual of correct motion sense carried by the language token – while the never-trained reverse instruction falls to *near-chance* (slightly negative, -0.19). The reverse split is genuinely negative on direction, but -0.19 is near-chance, not strong reliable anti-correlation; we do not claim the model reliably points the wrong way. Both are far below the in-dist 0.84 .

Findings (cont.)

2. **Accuracy–MAE inversion.** The seen-instruction task has *higher* absolute MAE (9.17 vs 8.40°) yet *better* within-5% (35.2% vs 19.5%) and a positive direction cosine. MAE and direction quality are decoupled: absolute error is governed by workspace geometry while the seen instruction still buys partial directional correctness. (Note: the per-task column is the source's "within-5%" metric, not the 5-degree threshold.)

Critically, the seen-instruction split being **worse on absolute MAE** (9.17 vs 8.40°) rules out "it failed only because the instruction was novel." The failure is a real scene/collection distribution shift, not merely unseen language – see §7.

5. Per-joint / kinematic structure

The same error structure appears in both conditions, scaled up under OOD
(`figs/perjoint\_indist.png`, `figs/perjoint\_ood.png`).

Joint	In-dist horizon-end MAE	OOD horizon-end MAE
j1	0.92°	6.32°
j2	0.86°	6.83°
j3	1.57°	10.29°
j4	1.85°	14.87°
j5	1.26°	6.84°
j6	1.38°	7.77°

- **j4 is the hardest joint and j3 second in both conditions**, with the base pair (j1, j2) cheapest as a group. The *worst* ranking (j4 > j3, then j6, j5) is preserved across conditions; the base pair stays cheapest but its internal order flips (in-dist cheapest is j2 then j1; OOD cheapest is j1 then j2), so the ranking is preserved as a structure, not byte-for-byte identical.

- The OOD blowup is largest exactly on the joints that carry the most predictive burden (j4 1.85 → 14.87°, j3 1.57 → 10.29°), while the base joints that move through a narrow, repeatable arc stay relatively cheap (j1 6.32, j5 6.84°). This is consistent with the workspace-prior reading: joints that barely move keep small absolute error; joints that must execute the actual reach/place motion are where tracking failure becomes visible. *(Caveat: per-joint MAE is in raw degrees and is not normalized by each joint's range of travel, so "worst in MAE" is not identical to "worst in normalized fidelity"; the all-negative OOD R² nonetheless confirms a genuine collapse on j3/j4.)*

- The preserved rank, plus j4 being worst in both, argues the OOD failure is a **uniform scaling of the same error structure** – not a new failure mode, a single pathological joint, or a unit/indexing bug.

6. Caveats & confounds

Findings (cont.)

- **The "bottle → box" OOD split shares its instruction string AND object with a trained multiobject task** (the training set contains 299/424 bottle→box episodes). Only the *recorded episodes* are held-out (a separate `farm_uf850_bottle` collection). It is therefore a **cross-collection / same-task shift**, not pure novel-task OOD. The clean, unambiguous novel-instruction evidence is the reverse "bottle → desk" split ($n = 348$, 8.40° , $\text{dir-cos} -0.19$), which fails comparably. Because the seen-instruction split is in fact *worse* on absolute MAE (9.17 vs 8.40°), the conclusion holds regardless: the failure is not explained by unseen language. (On *direction*, the seen split is slightly less collapsed, $+0.13$ vs -0.19 , but both are far below the in-dist 0.84 .)
- **Teacher-forced ≠ closed-loop.** Every prediction is conditioned on the real recorded observation, so this measures single-shot prediction fidelity, not on-arm task success, grasp completion, or recovery. No success-rate claim can be made from these numbers. **Closed-loop OOD** would almost certainly be *worse*: with direction cosine at chance, the policy doesn't know which way to move from a true state, so once its own wrong actions push the arm off-manifold the errors would compound without ground-truth re-grounding. The open-loop 8.82° is therefore an **optimistic floor** on OOD badness – this is an inference, not a measured number.
- **Workspace-prior effect on absolute MAE.** Absolute-joint MAE is flattered by the workspace prior on *both* conditions, so the $6.7\times$ ratio *understates* the collapse. The prior-free metrics (displacement Pearson $0.940 \rightarrow 0.079$, direction cosine $0.84 \rightarrow \text{chance}$) are the load-bearing generalization evidence. We did not include an explicit static-pose baseline MAE to quantify exactly how much of 8.82° is prior vs. residual signal, though the all-negative R^2 implies the model is below that baseline on displacement.
- **Normalization is valid, not a handicap.** Both conditions use the identical checkpoint (`.../5999`) and config (`pi05_farm_multiobject_gse`), so `create_trained_policy` loads the **same** norm-stats (computed from multiobject training) for both – exactly what happens at deploy. Both datasets are the same UF850, same 7-DoF absolute joint space, same 30 fps, same base+wrist cameras. Normalizing OOD data with training stats is correct, not an unfair inflation. Prompts are read verbatim (`prompt_from_task=True`) with augmentation disabled at inference, so there is no prompt mismatch – the "bottle → box" OOD instruction is byte-identical to a trained one.
- **In-distribution is the model's OWN training set** (FIT, with likely memorization given $r = 0.94$ and the flat horizon curve), so it is an upper bound on fit, not a held-out same-distribution validation score. A held-out in-domain split would be a stronger anchor.
- **Sample asymmetry / imbalance.** Matched seed (0), frame count (768), episodes (64), window `[0.25, 0.75]`, and horizon (10) across conditions. But in-dist mixes 4 tasks (bottle-dominated, 528/768; duck $n=24$ and hat $n=84$ are thin/noisier) while OOD is 2 bottle tasks. Since bottle is the in-dist *strongest* task (1.24°), this if anything makes the in-dist baseline tougher to beat, not easier.
- **Per-project context:** prior diagnosis attributed live failures partly to a deploy-room/camera domain shift. The seen-instruction split being worse than the novel reverse task is consistent with a scene/collection domain gap between collections – i.e., part of "OOD" here is environment shift, not purely task/semantic generalization.

7. Bottom line

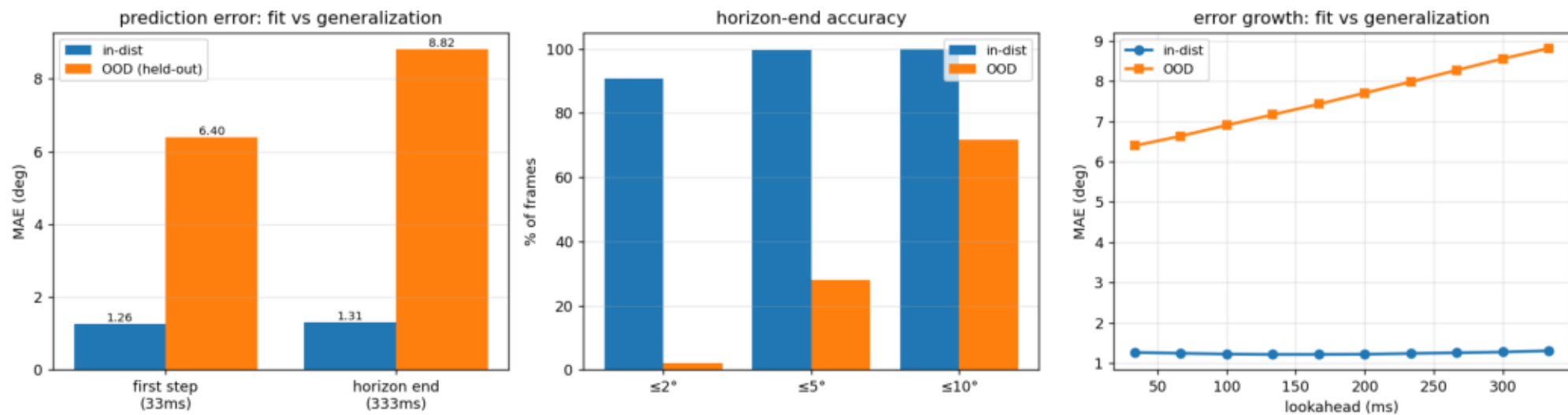
On its 4-task multiobject training set, the GSE full fine-tune of $\pi 0.5$ fits the demonstrated

Findings (cont.)

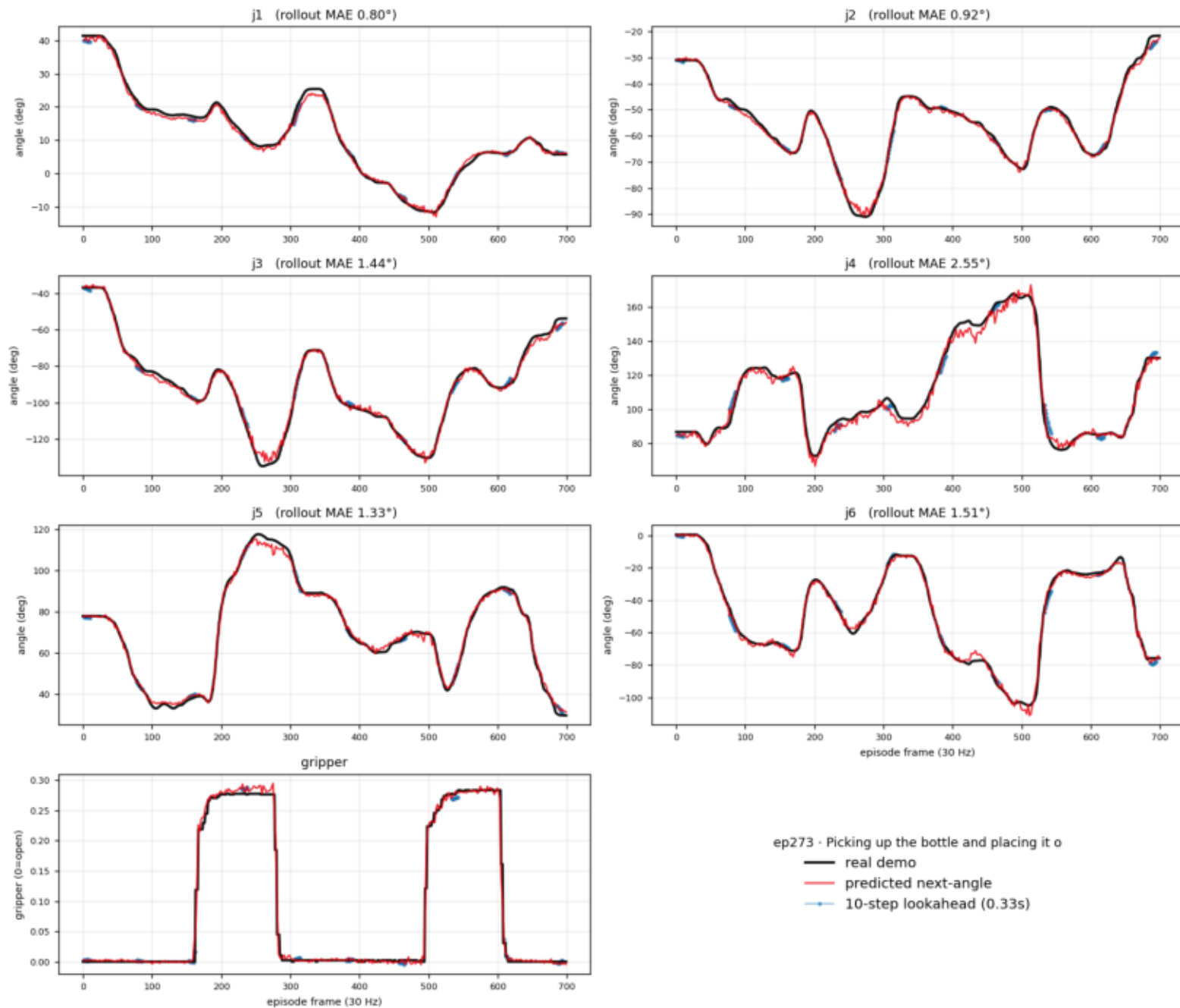
trajectories tightly – $\sim 1.3^\circ$ non-compounding prediction error out to 333 ms, displacement $r = 0.940$, direction cosine 0.84, gripper 99.0% – but on held-out episodes its motion prediction collapses to chance: displacement $r = 0.079$, direction cosine -0.013 , all-negative per-joint R^2 , and a now-growing error-vs-horizon curve. The bounded absolute MAE (8.82° , 71.7% within 10°) is best explained by the model staying inside the learned, low-dimensional joint workspace while retaining **no transferable, scene-conditioned motion policy**. The familiar instruction string buys only a faint directional prior ($+0.13$ seen vs -0.19 never-trained), confirming the model leans on memorized training-episode associations rather than generalizing from the visual scene.

This corroborates the project's prior findings: a small-multi-task full fine-tune fits its training manifold extremely tightly but learns a **workspace prior**, not a generalizable visuomotor skill. When held-out observations fall just off that manifold, the policy snaps to a plausible-but-scene-blind pose – which is exactly why live OOD deployment failed, and why the chance-level direction cosine (the prior-free signal), not the deceptively bounded MAE, is the number that would govern closed-loop behavior. The aggregate side-by-side is summarized in `\figs/compare_indist_vs_ood.png``.

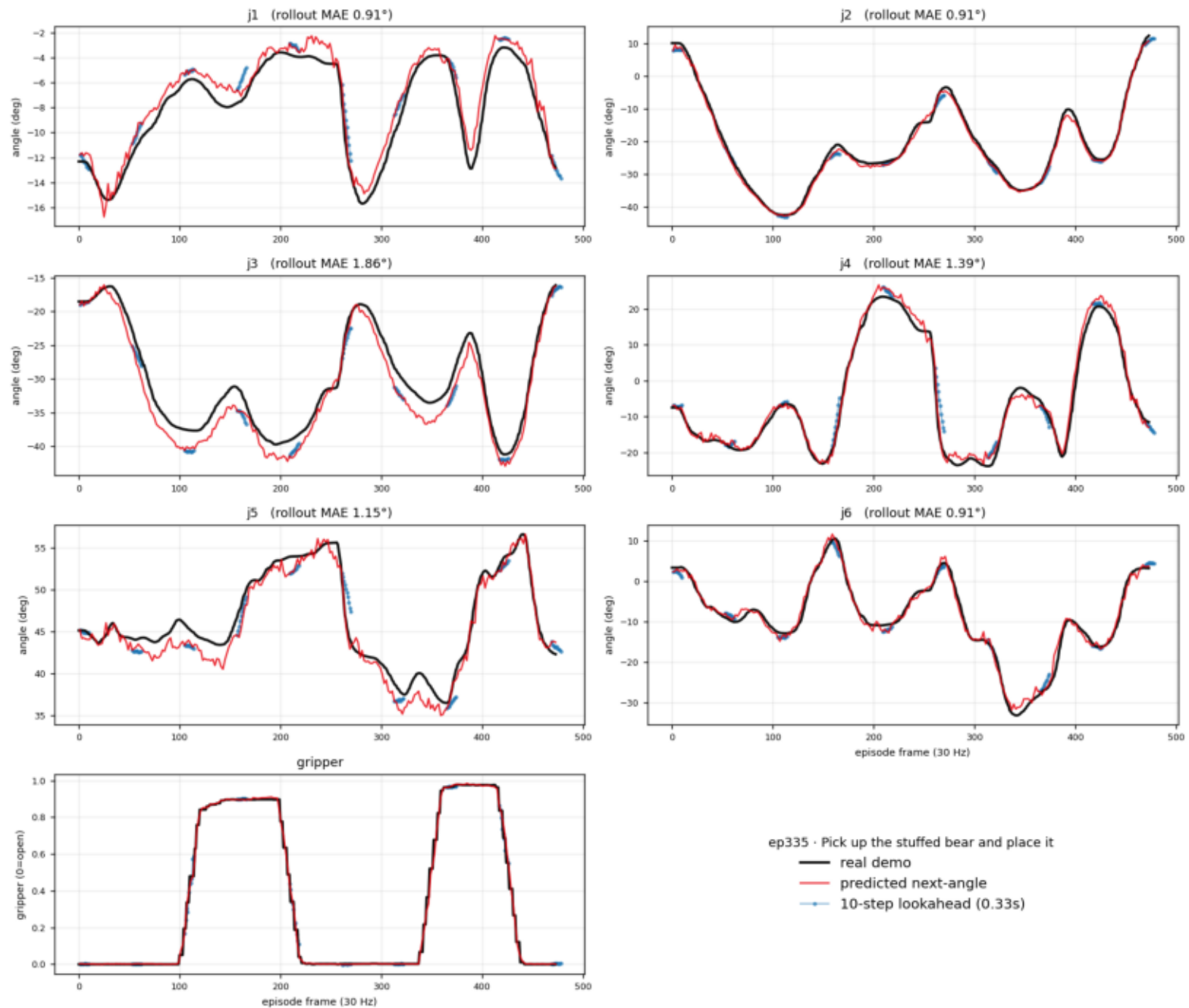
Multiobject GSE — in-distribution fit vs out-of-distribution generalization

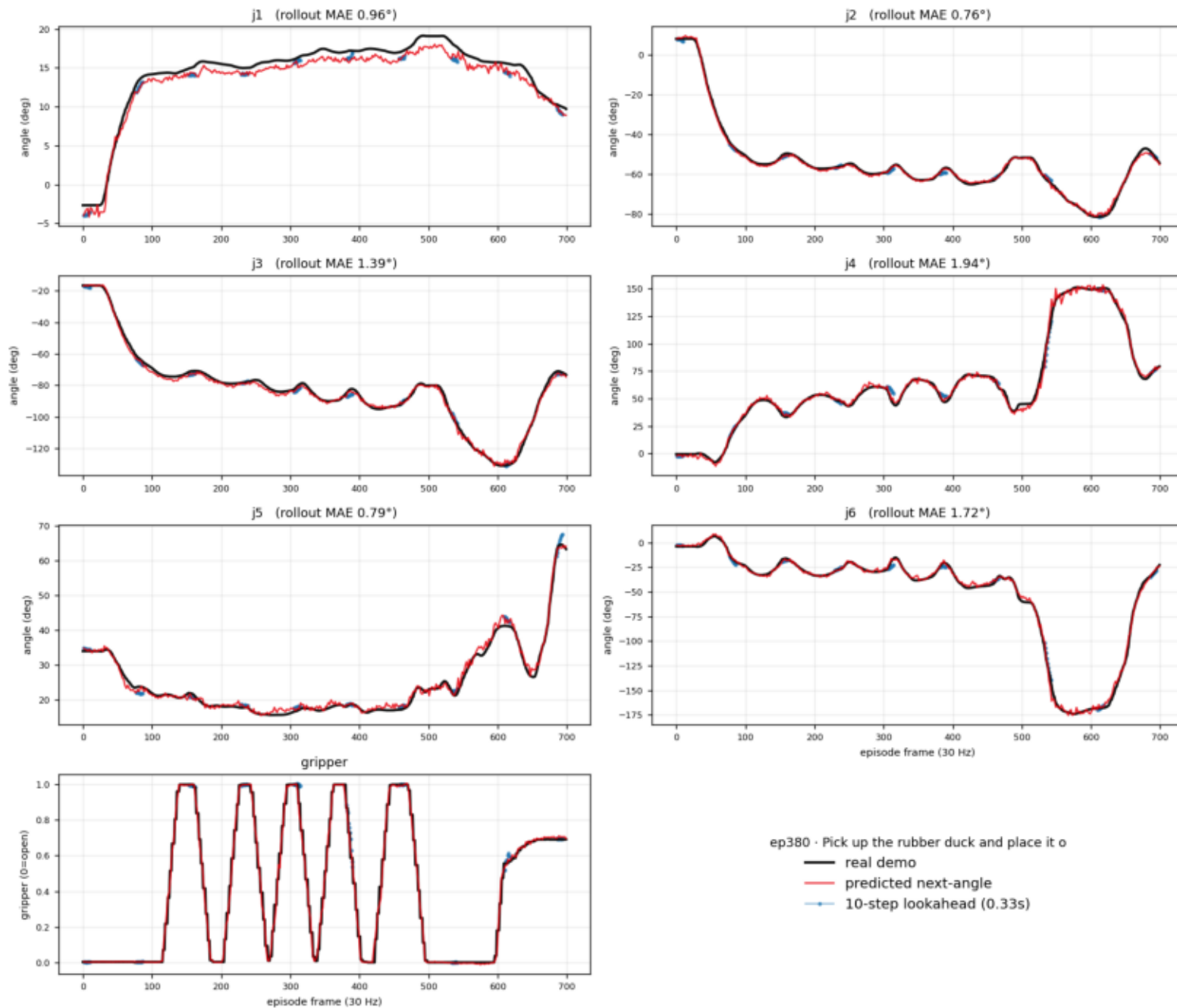


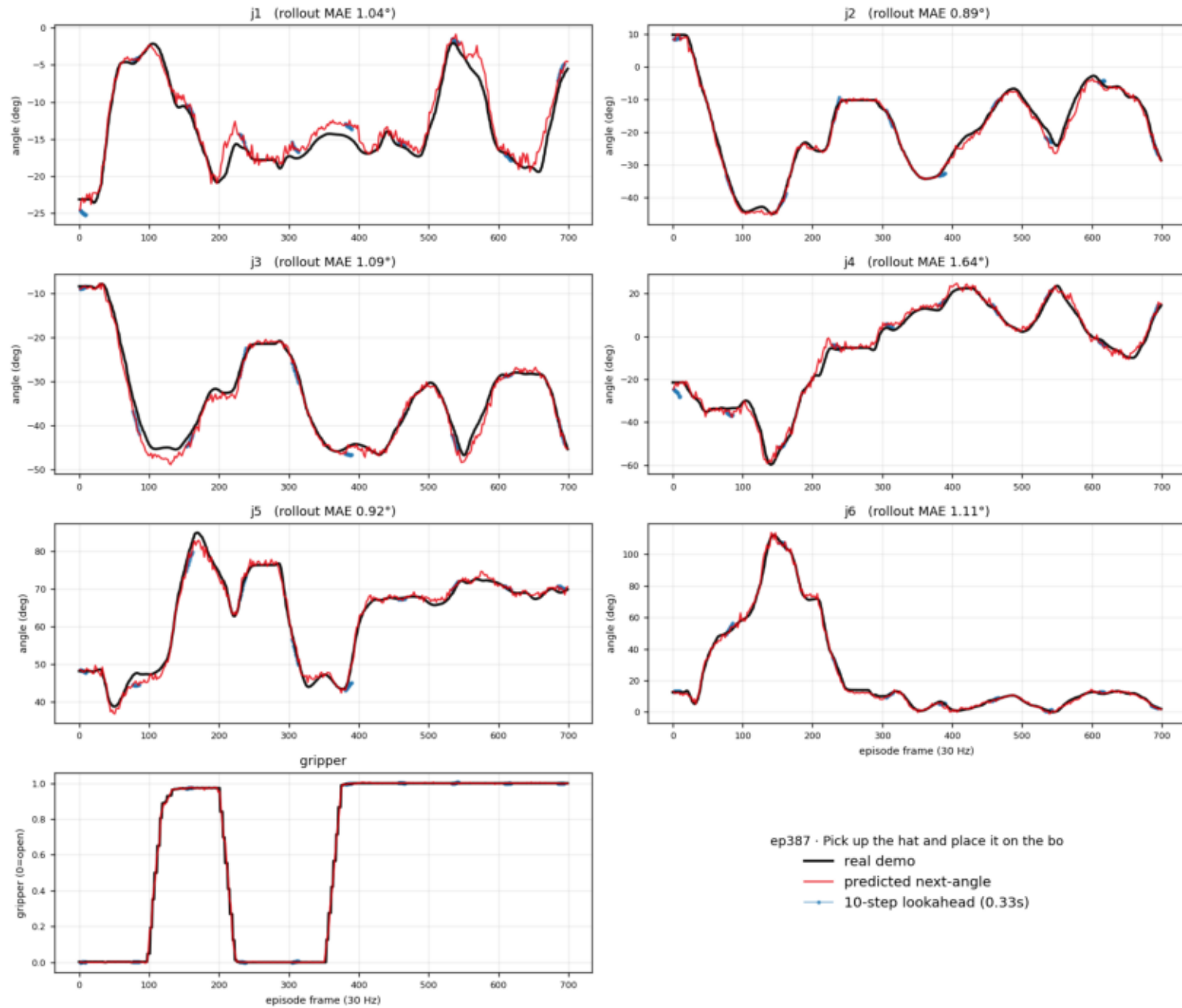
Headline: in-distribution fit vs out-of-distribution generalization.



IN-DIST · bottle→box: commanded next-angle (red) vs real demo (black), per joint + gripper.

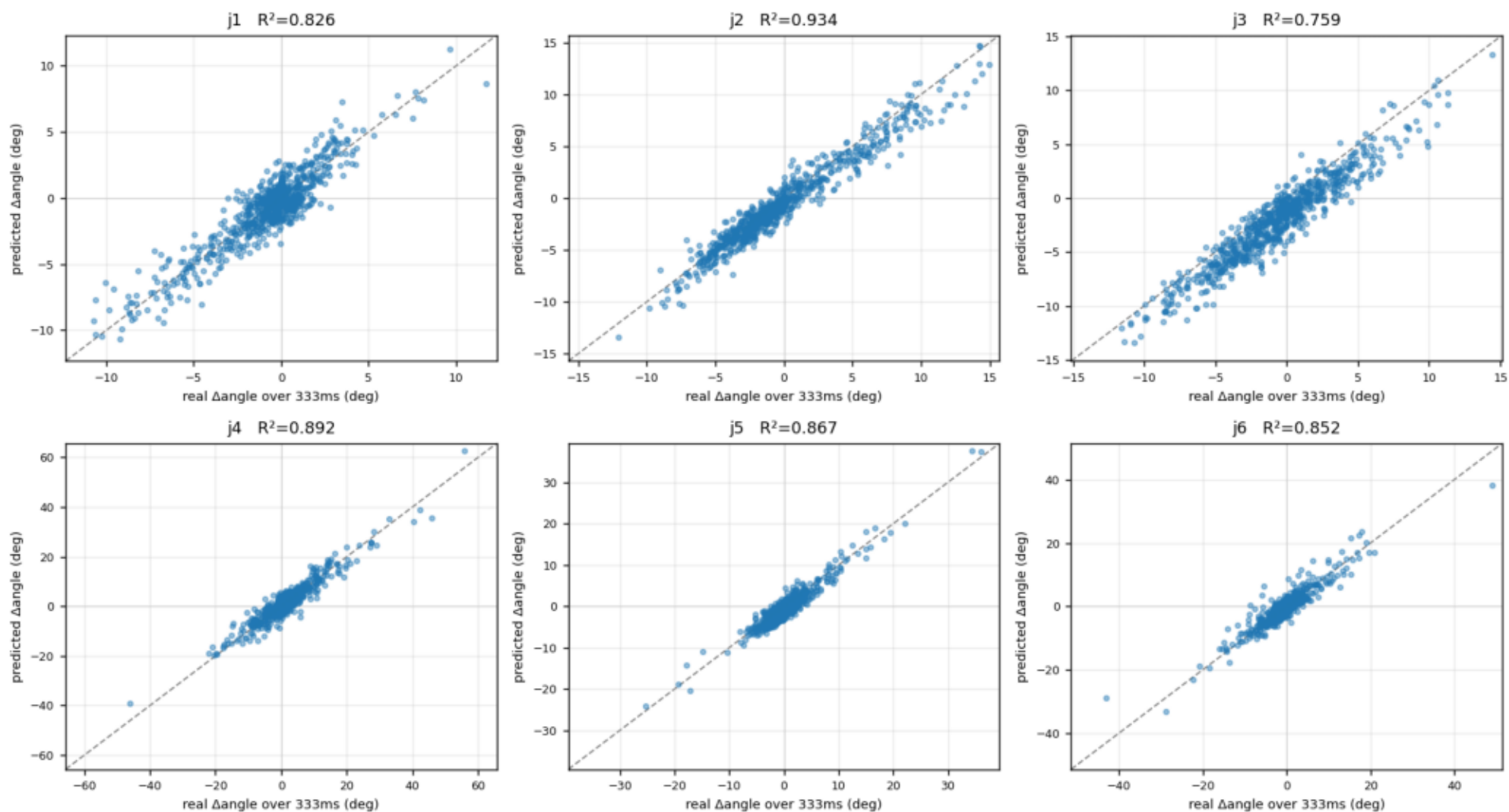






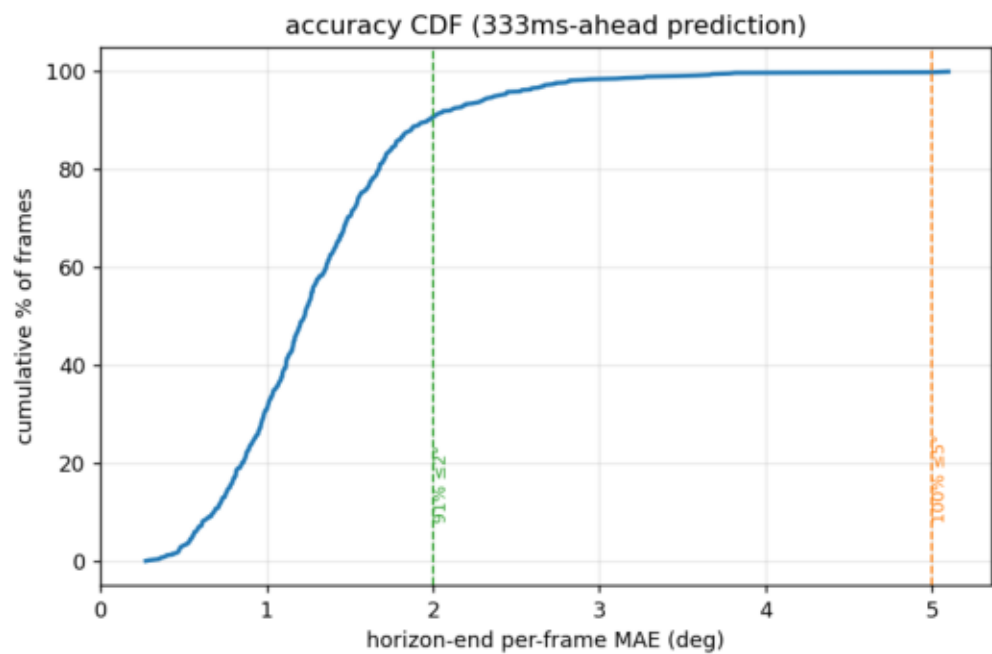
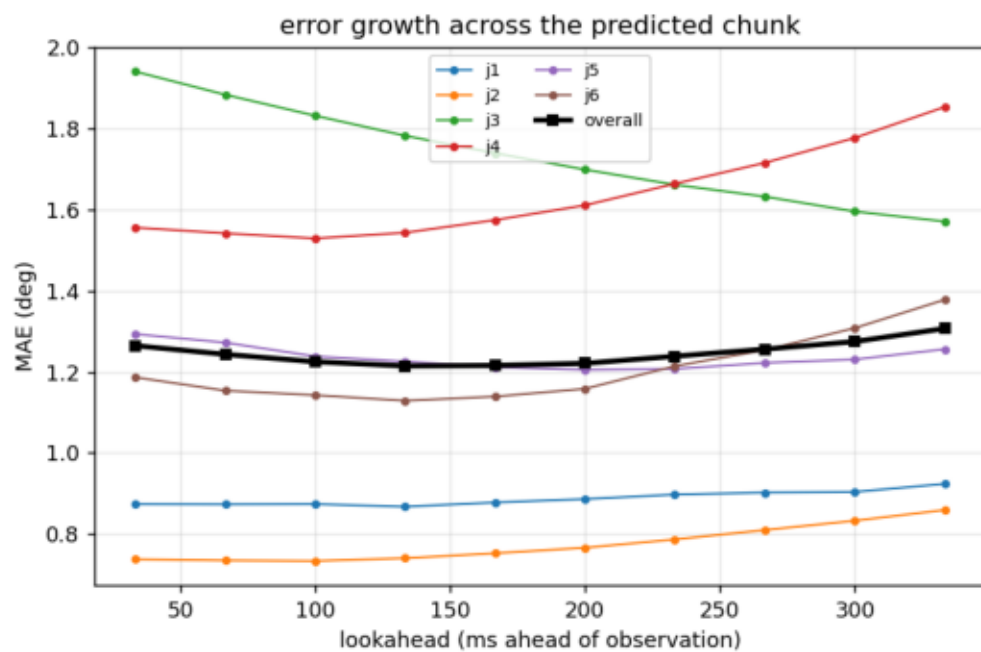
In-distribution (multiobject training set)

predicted vs real joint displacement over the 10-step horizon (Pearson $r=0.940$) — points on the diagonal = the model commands the demonstrated motor



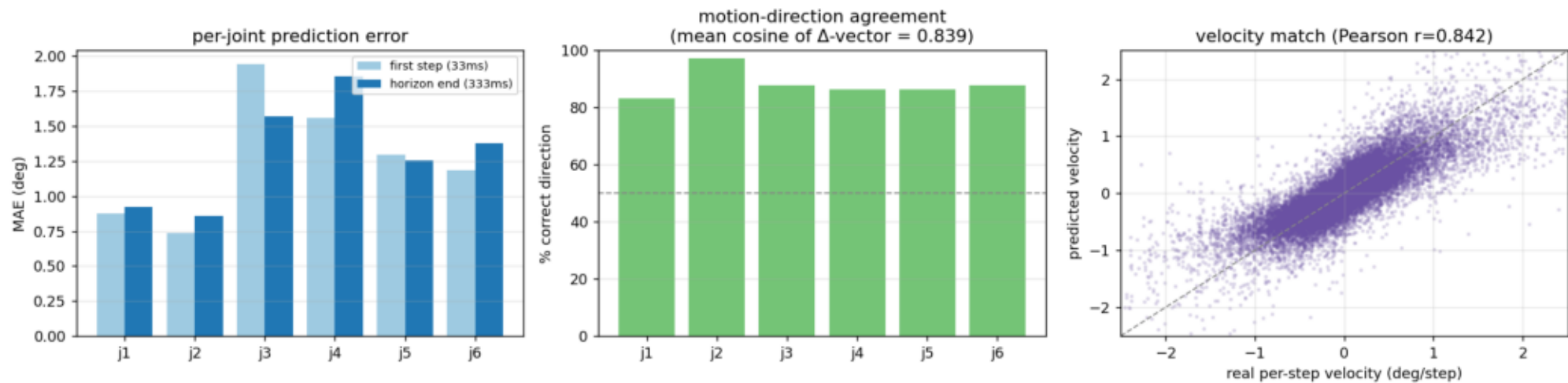
IN-DIST · predicted vs real joint displacement over the 333ms horizon (diagonal = perfect).

In-distribution (multiobject training set)



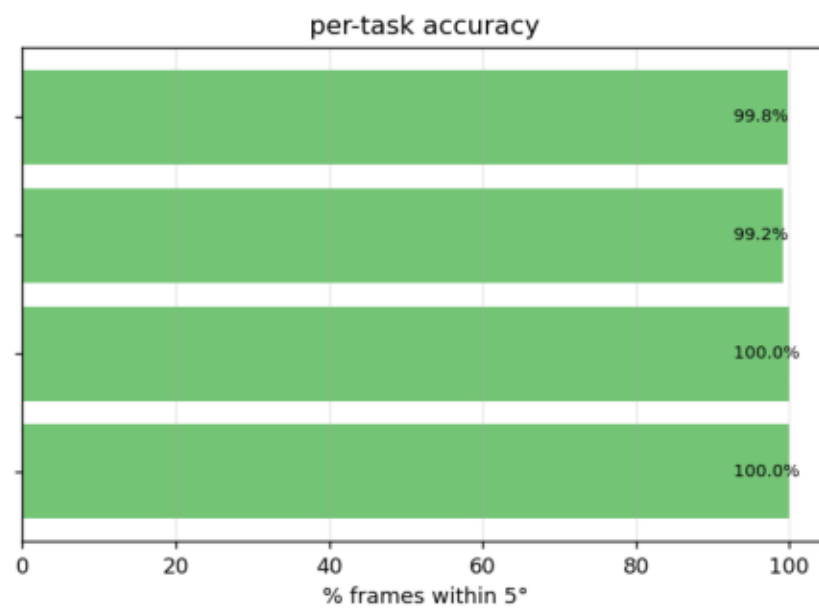
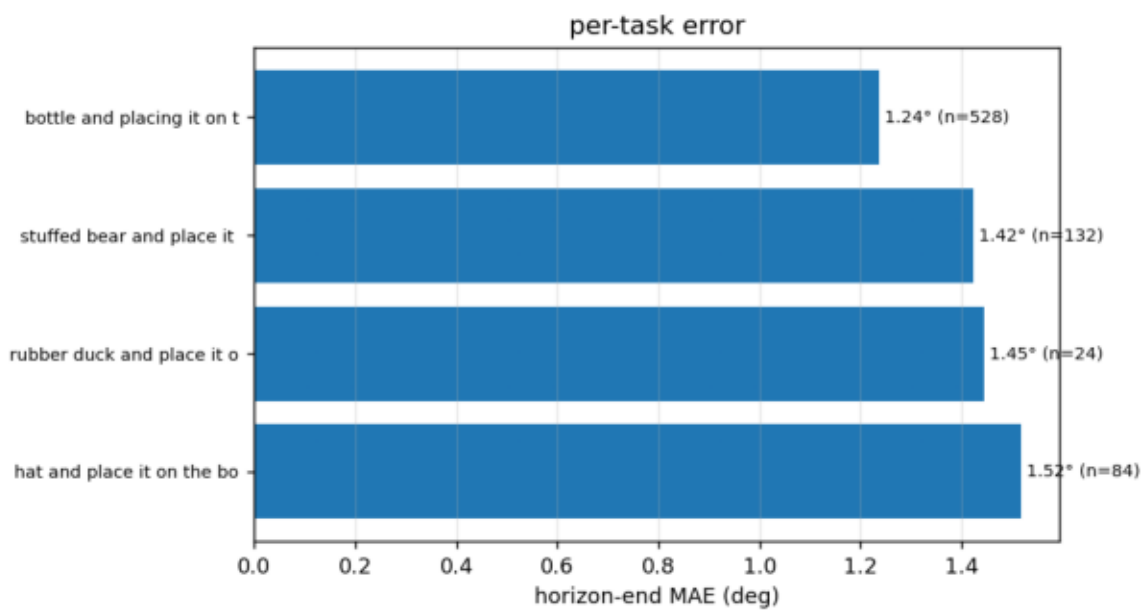
IN-DIST · error growth across the chunk + horizon-end accuracy CDF.

In-distribution (multiobject training set)

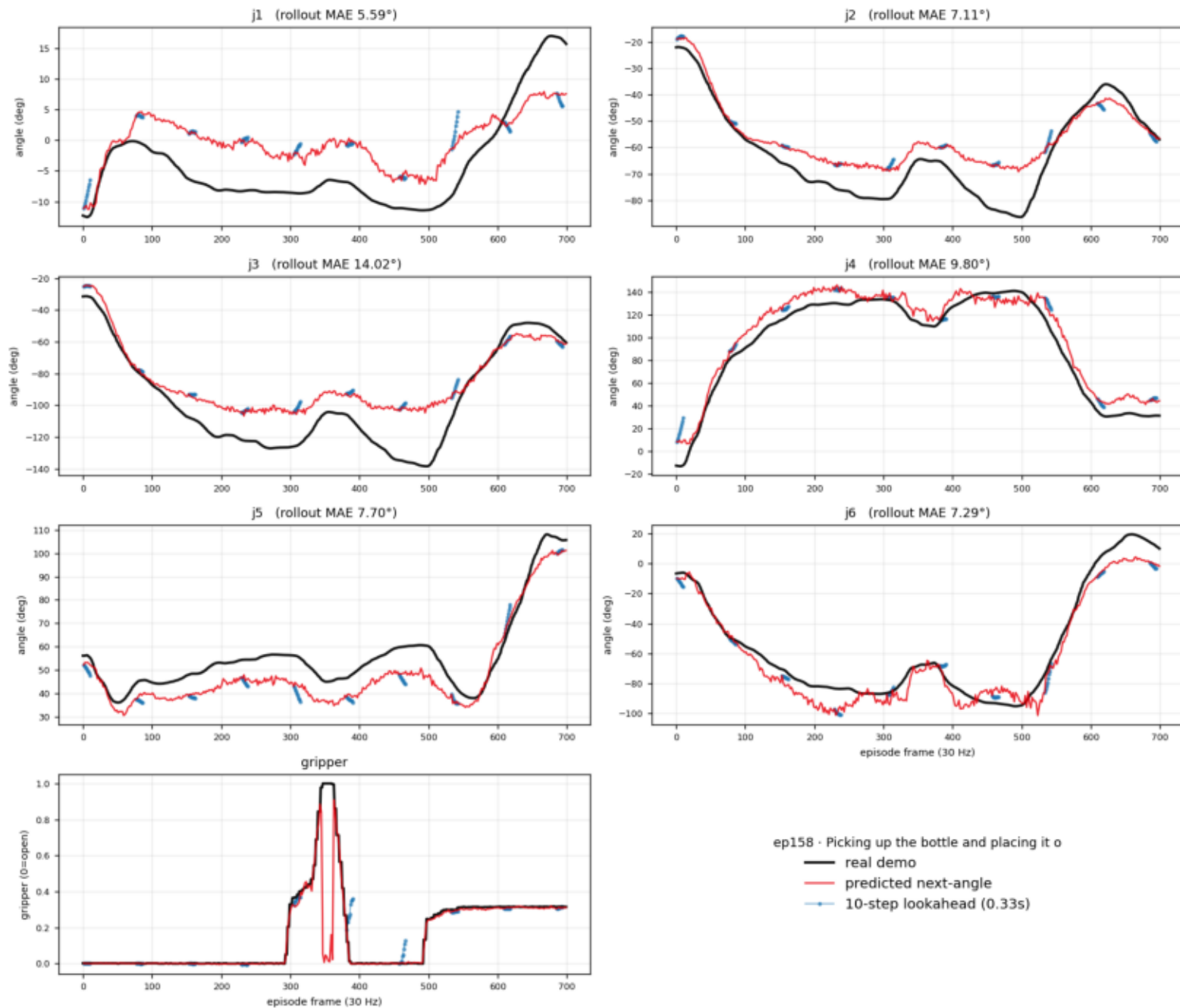


IN-DIST · per-joint error, motion-direction agreement, velocity match.

In-distribution (multiobject training set)

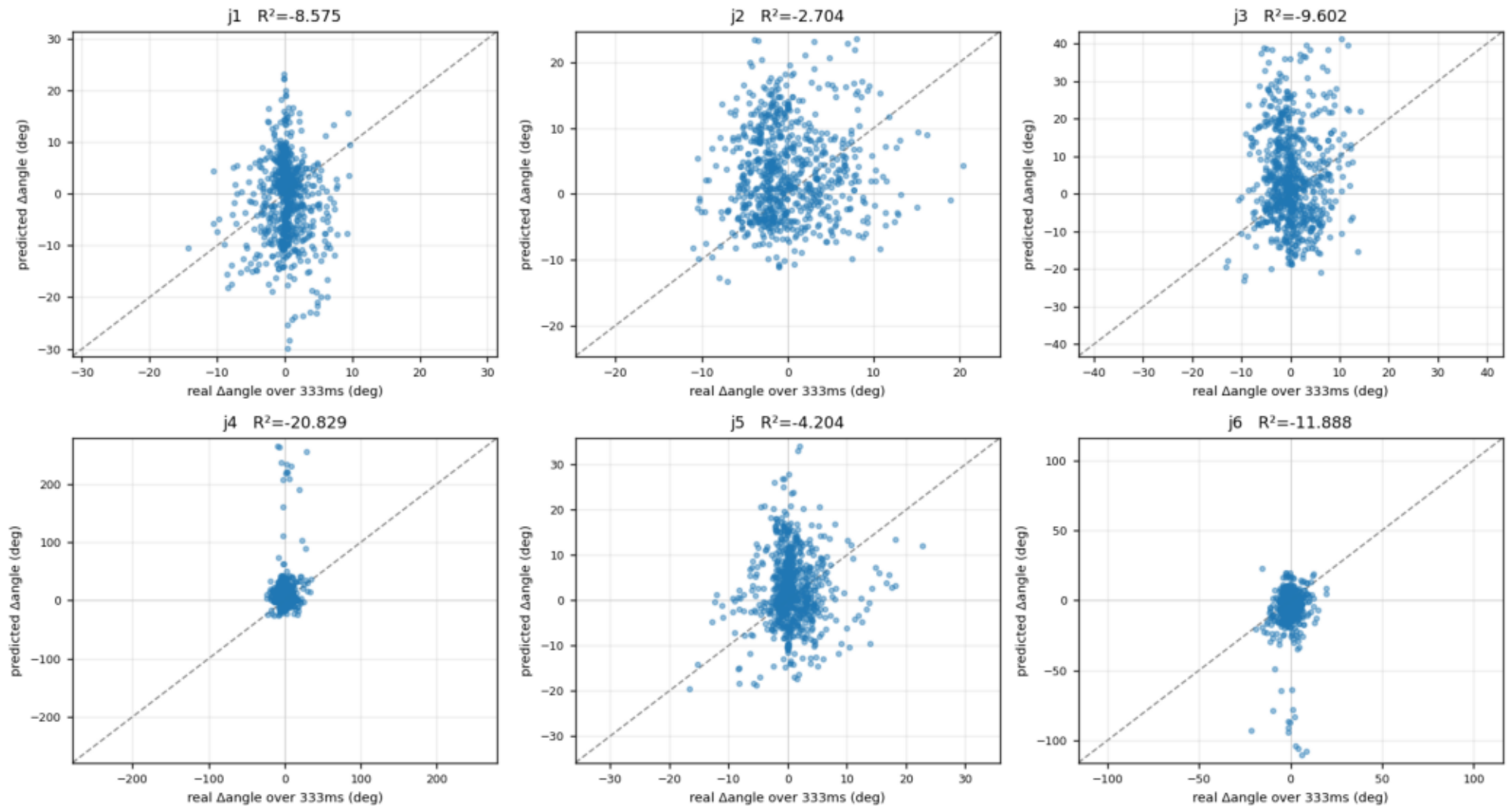


IN-DIST · per-task error and accuracy.



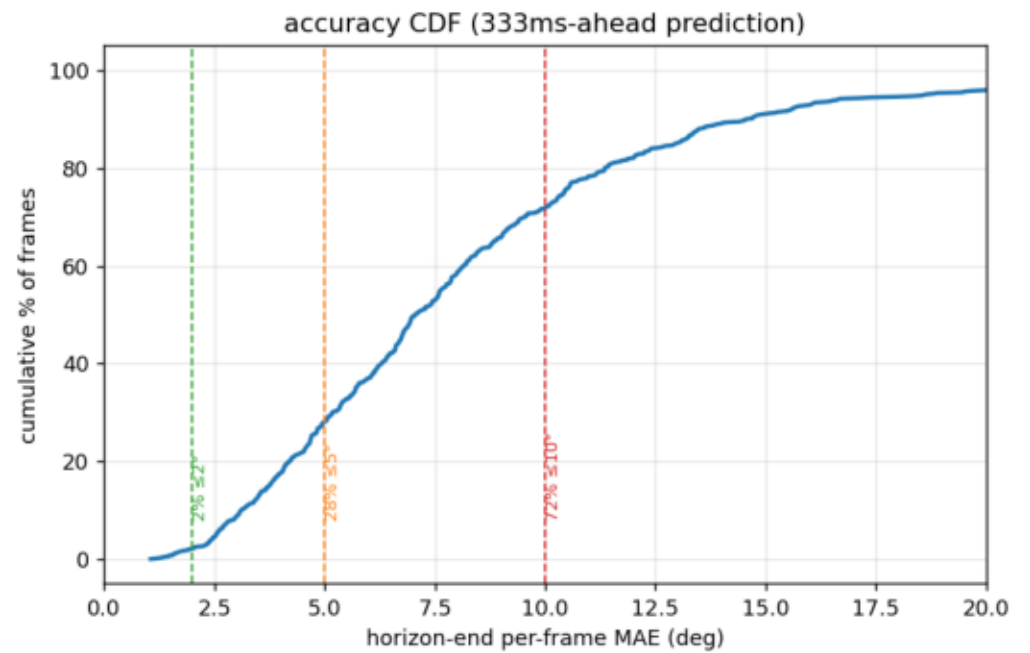
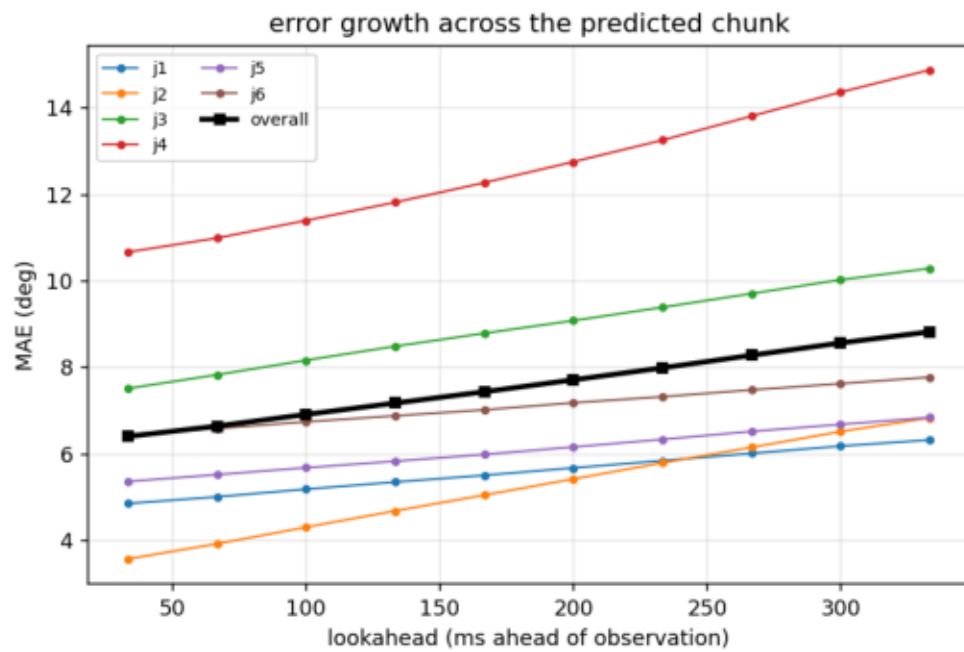
Out-of-distribution (held-out farm_uf850_bottle)

predicted vs real joint displacement over the 10-step horizon (Pearson $r=0.079$) — points on the diagonal = the model commands the demonstrated motor



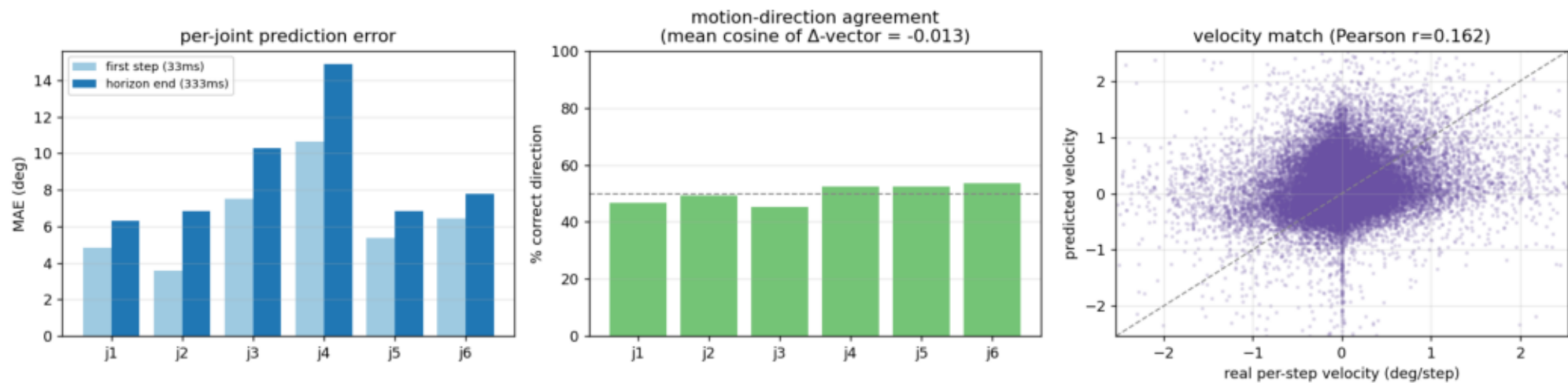
OOD · predicted vs real displacement on held-out data.

Out-of-distribution (held-out farm_uf850_bottle)



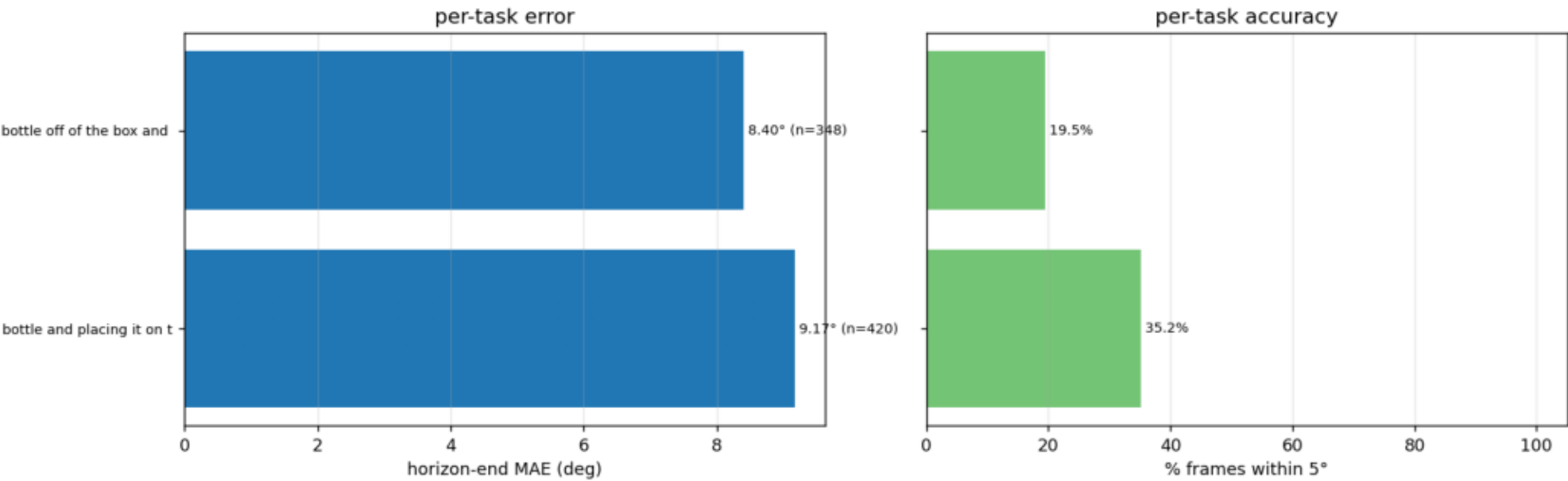
OOD · error growth + accuracy CDF on held-out data.

Out-of-distribution (held-out farm_uf850_bottle)



OOD · per-joint error, direction, velocity on held-out data.

Out-of-distribution (held-out farm_uf850_bottle)



OOD · per-task error (separates seen-instruction held-out episodes from the never-trained reverse task).